

# What Brevo now does for the shop

Everything built on the free Brevo tier — five customer emails, list management, bounce handling, abandoned-cart recovery and delivery SMS — and what each one actually buys you.

IMPLEMENTATION RECORD · BRANCH `CLAUDE/ANALYZE-UI-DESIGN-5KPX1`

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5

CUSTOMER EMAILS

4

BREVO CAPABILITIES

543

TESTS PASSING

1

VERCEL FUNCTION

## The shape of it

Brevo does two separate jobs here, and keeping them separate is deliberate.

### Transactional — the REST API

Messages a customer's own action triggers: welcome, order confirmed, dispatched, out for delivery, basket reminder. These must send instantly and must never be blocked by a marketing tool being down.

### Campaigns — the SMTP relay

Newsletters and promotions, composed by a person and sent to a list. Needs segmentation, open tracking and an unsubscribe link that transactional mail must *not* carry.

Both hand off to the same Brevo account on purpose. One sending domain, one DKIM signature, one reputation. Splitting them across two providers is what gets order confirmations filed as spam because a promotional blast annoyed people last week.

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## 1 · The five customer emails

All built from the storefront's own design system — stone neutrals against the gold accent, the LeHart wordmark, the same 14px card. Not a second visual language invented for email.

| EMAIL                     | FIRES WHEN                 | WHAT IT DOES FOR YOU                                                                                   |
|---------------------------|----------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Welcome</b>            | Newsletter signup          | Delivers the WELCOME20 code and sets expectations, so the first campaign is not a cold approach.       |
| <b>Order confirmation</b> | Order written successfully | Itemised receipt with server-computed totals. The single most-opened email a shop sends.               |
| <b>Dispatched</b>         | Staff marks it dispatched  | Carries the courier and tracking number, which is what removes the "where is my phone" support ticket. |
| <b>Out for delivery</b>   | Staff marks it out         | Tells the customer to be in. Fewer failed deliveries, fewer redelivery costs.                          |
| <b>Abandoned cart</b>     | Hourly sweep, 4h after     | The only one that directly recovers revenue.                                                           |

Every message ships a plain-text part alongside the HTML, a hidden preheader so the inbox preview is not just the word "LeHart", and a four-stop delivery tracker on the order emails. Everything customer-supplied is HTML-escaped.

#### Why tables and not modern CSS

Outlook renders through Word's HTML engine, which supports neither flexbox nor grid. A div-based layout collapses into a single column there. Every rule is inlined on the element it styles, because email has no stylesheet and no cascade worth relying on.

## 2 • Bounce and complaint handling the important one

Before this, nothing in the shop noticed when an email bounced or someone marked it as spam.

That is a slow-motion failure. Keep mailing a dead address and providers read it as a list you do not maintain. Collect spam complaints and they stop trusting the domain for *everything* — order confirmations included. You do not see it happen; you just watch delivery quietly decline.

| BREVO EVENT                                                        | ACTION TAKEN                                                                    |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------|
| hard_bounce, blocked, spam, complaint, unsubscribed, invalid_email | Suppressed in <b>all three</b> systems at once — Firestore, Brevo and Listmonk. |
| delivered, opened, click, soft_bounce, deferred                    | Logged only. Open and click rates are your read on whether the templates work.  |

A soft bounce never suppresses — it is transient, and a temporarily full mailbox is not a dead customer.

### Brevo does not sign its webhooks

There is no HMAC to verify, so the secret in the URL is the entire authentication boundary. It is compared in constant time, and with the secret unset the route returns 404 rather than defaulting open — an endpoint that suppresses addresses, left open, is a denial-of-service against your own customers. Rotate it if the URL ever leaks.

## 3 • Contacts and campaigns

Newsletter signups now land in Brevo contacts as well as Listmonk. Both are *projections*; Firestore stays the source of truth, holding the consent record — timestamp, source, policy version, truncated IP — that makes the list lawfully mailable under UK GDPR and PECR.

That design means the decision you have not made yet stays open: whether campaigns go out from Brevo (no server to run) or Listmonk (no per-contact ceiling) can change later without touching the signup path at all.

Suppression sets Brevo's `emailBlacklisted` flag rather than removing the contact from a list, because a contact merely taken off a list can still be picked up by an automation.

## 4 • Abandoned cart recovery

The only piece here that directly makes money.

### 1 Checkout starts

The basket is recorded the moment an email address is known — the earliest point recovery is possible at all.

### 2 Four hours pass

Someone who wandered off for ten minutes is still shopping. An email then is an interruption, not a service.

### 3 One reminder sends

Stamped before the send, so a crash mid-batch cannot produce a second email. Losing a reminder is far cheaper than sending two.

### 4 After 48 hours, nothing

A three-day-old basket is a different decision, and mailing about it reads as surveillance.

### Deliberately no discount code

Handing one to everyone who hesitates teaches customers to abandon on purpose, and costs margin on the ones who were coming back anyway. For refurbished phones the doubt being answered is "is this thing any good" — so the email carries the warranty, the 60-point inspection and the returns window instead.

The stored basket is **display-only**. Real prices are recomputed server-side when the order is actually placed. Quoting a browser-supplied total in a recovery email would reintroduce exactly the price-tampering hole the order route was rewritten to close.

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## 5 • Delivery SMS

Wired to the out-for-delivery message only — the one notification where nobody is checking their inbox. Opt-in per call, never automatic.

### **SMS is not part of the free tier**

It bills prepaid credits separately from email. With no sender configured every send is a silent no-op, so a shop that has not deliberately turned SMS on can never accidentally spend. Messages truncate at 160 characters — one GSM-7 segment — so a runaway template cannot quietly cost several credits per send.

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## The 300-a-day ceiling

The free tier's daily allowance is shared between transactional and marketing mail. For current volumes that is comfortable, but the failure mode matters: a large campaign or a recovery backlog could in principle crowd out an order confirmation, which is the worst thing that can happen to a shop's email.

The abandoned-cart sweep is therefore capped at 25 messages an hour by default, specifically so a backlog can never eat the quota the order emails depend on. Check the remaining allowance in the dashboard before your first large campaign.

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## Everything fails softly

Every integration here is a logged no-op when its keys are missing, and none of them can fail a customer's action:

- An order is durably written before the confirmation is attempted. A dead mail provider must never turn a successful purchase into an error the customer would reasonably retry — that is how you get two orders and one angry email.
- A subscription is recorded before the welcome email or either list sync runs.
- A failed SMS never affects the email that already went, or the order status.

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## What is left to do

| STEP                                 | WHY                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Rotate the API key</b>            | The key shared in chat should be treated as compromised. Generate a new one and add it to Vercel only. |
| <b>Verify a sender domain</b>        | Until DKIM and SPF are set on the sending domain, everything is far more likely to land in spam.       |
| <b>Set the environment variables</b> | Every feature here stays dormant until its keys reach the deployment.                                  |
| <b>Point the webhook at the site</b> | Bounce handling only starts working once Brevo knows where to post.                                    |
| <b>Merge to the deployed branch</b>  | This work sits on a branch Vercel does not build. Until it is merged, none of it is live.              |